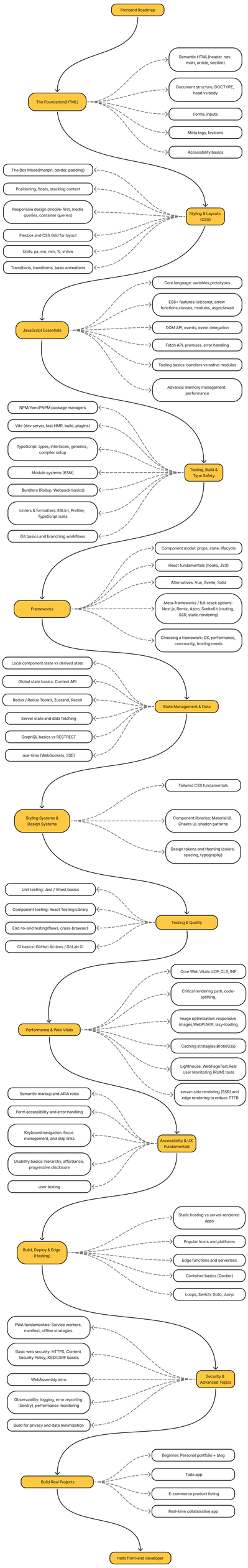
Frontend Roadmap
The Foundation(HTML)
Semantic HTML(header, nav, main, article, section)
Document structure, DOCTYPE, head vs body
Forms, inputs
Meta tags, favicons
Accessibility basics
Styling & Layouts (CSS)
The Box Model(margin, border, padding)
Positioning, floats, stacking context
Responsive design (mobile-first, media queries, container queries)
Flexbox and CSS Grid for layout
Units: px, em, rem, %, vh/vw
Transitions, transforms, basic animations
JavaScript Essentials
Core language: variables,prototypes
ES6+ features: let/const, arrow functions,classes, modules, async/await
DOM API, events, event delegation
Fetch API, promises, error handling
Tooling basics: bundlers vs native modules
Advance: Memory management, performance
Tooling, Build & Type Safety
NPM/Yarn/PNPM package managers
Vite (dev server, fast HMR, build, plugins)
TypeScript: types, interfaces, generics, compiler setup
Module systems (ESM)
Bundlers (Rollup, Webpack basics)
Linters & formatters: ESLint, Prettier, TypeScript rules
Git basics and branching workflows
Frameworks
Component model: props, state, lifecycle
React fundamentals (hooks, JSX)
Alternatives: Vue, Svelte, Solid
Meta-frameworks / full-stack options: Next.js, Remix, Astro, SvelteKit (routing, SSR, static rendering)
Choosing a framework: DX, performance, community, hosting needs
State Management & Data
Local component state vs derived state
Global state basics: Context API
Redux / Redux Toolkit, Zustand, Recoil
Server state and data fetching
GraphQL basics vs RESTREST
real-time (WebSockets, SSE)
Styling Systems & Design Systems
Tailwind CSS fundamentals
Component libraries: Material UI, Chakra UI, shadcn patterns
Design tokens and theming (colors, spacing, typography)
Testing & Quality
Unit testing: Jest / Vitest basics
Component testing: React Testing Library
End-to-end testing(flows, cross-browser)
CI basics: GitHub Actions / GitLab CI
Performance & Web Vitals
Core Web Vitals: LCP, CLS, INP
Critical rendering path, code-splitting,
Image optimization: responsive images,WebP/AVIF, lazy-loading
Caching strategies,Brotli/Gzip
Lighthouse, WebPageTest,Real User Monitoring (RUM) tools
server-side rendering (SSR) and edge rendering to reduce TTFB
Accessibility & UX Fundamentals
Semantic markup and ARIA roles
Form accessibility and error handling
Keyboard navigation, focus management, and skip links
Usability basics: hierarchy, affordance, progressive disclosure
user testing
Build, Deploy & Edge (Hosting)
Static hosting vs server-rendered apps
Popular hosts and platforms
Edge functions and serverless
Container basics (Docker)
Loops, Switch, Goto, Jump
Security & Advanced Topics
PWA fundamentals: Service workers, manifest, offline strategies.
Basic web security: HTTPS, Content Security Policy, XSS/CSRF basics
WebAssembly intro
Observability: logging, error reporting (Sentry), performance monitoring
Build for privacy and data minimization
Build Real Projects
Beginner: Personal portfolio + blog
Todo app
E-commerce product listing
Real-time collaborative app
hello front-end developer